

1(a)

The variable X has the distribution $B(50, \frac{1}{6})$. The probabilities $P(X = r)$ for $r = 0$ to 50 are given by the terms of the expansion of $(a+b)^n$ for specific values of a , b and n .

State the values of a , b and n .

[1]

- (b) A student has an ordinary 6-sided dice. They suspect that it is biased so that it shows a 2 on fewer throws than it would if it were fair. In order to test the suspicion the dice is thrown 50 times and the number of 2s is noted. The student then carries out a hypothesis test at the 5% significance level.

Write down suitable hypotheses for the test.

[2]

- (c) Determine the rejection region for the test, showing the values of any relevant probabilities.

[4]

- 2(a) Some packets of a certain kind of biscuit contain a free gift. The manufacturer claims that the proportion of packets containing a free gift is 1 in 4. Marisa suspects that this claim is not true, and that the true proportion is less than 1 in 4. She chooses 20 packets at random and finds that exactly 1 contains the free gift.

Use a binomial model to test the manufacturer's claim, at the 2.5% significance level.

[7]

- (b) The packets are packed in boxes, with each box containing 40 packets. Marisa chooses three boxes at random and finds that one box contains 19 packets with the free gift and the other two boxes contain no packets with the free gift.

Give a reason why this suggests that the binomial model used in part (a) may not be appropriate.

[1]

- 3(a) It is known that, under the standard treatment for a certain disease, 9.7% of patients with the disease experience side effects within one year.

In a trial of a new treatment, a random sample of 450 patients with this disease was selected and the number X who experienced side effects within one year was noted.

State one assumption needed in order to use a binomial model for X .

[1]

- (b) It was found that 51 of the 450 patients experienced side effects within one year.

Test, at the 10% significance level, whether the proportion of patients experiencing side effects within one year is greater under the new treatment than under the standard treatment.

[7]

- 4 A bag contains 100 black discs and 200 white discs. Paula takes five discs at random, without replacement. She notes the number X of these discs that are black.

(a) Find $P(X = 3)$.

[2]

Paula decides to use the binomial distribution as a model for the distribution of X .

(b) Explain why this model will give probabilities that are approximately, but not exactly, correct.

[3]

- (c) Paula uses the binomial model to find an approximate value for $P(X = 3)$. Calculate the percentage by which her answer will differ from the answer in part (b). [2]

Paula now assumes that the binomial distribution is a good model for X . She uses a computer simulation to generate 1000 values of X . The number of times that $X = 3$ occurs is denoted by Y .

- (d) Calculate estimates of the limits between which two thirds of the values of Y will lie. [3]

5 (a) André throws a fair six-sided dice 30 times. The number of throws on which the score is six is denoted by X .

(i) State a suitable model for X , including the values of any parameters. [1]

(ii) Find $P(X = 9)$. [1]

(iii) Find $P(X \geq 9)$. [2]

(b) André has another six-sided dice. He suspects that this dice is biased so that it is more likely to show a six than if it were fair. He throws the dice 30 times and it shows a six on 9 throws. Test at the 5% significance level whether André's suspicion is justified. [6]

- 6 (a) Write down and simplify the first four terms in the expansion of $(x + y)^7$.
Give your answer in ascending powers of x .

[2]

- (b) Given that the terms in $x^2 y^5$ and $x^3 y^4$ in this expansion are equal, find the value of $\frac{x}{y}$.

[2]

- (c) A hospital consultant has seven appointments every day. The number of these appointments which start late on a randomly chosen day is denoted by L . The variable L is modelled by the distribution $B\left(7, \frac{3}{8}\right)$.
Show that, in this model, the hospital consultant is equally likely to have two appointments start late or three appointments start late.

[3]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1	a		$n = 50; a = \frac{5}{6}, b = \frac{1}{6}$ or $a = \frac{1}{6}, b = \frac{5}{6},$	B1 [1]	or $\left(\frac{5}{6} + \frac{1}{6}\right)^{50}$ or $\left(\frac{1}{6} + \frac{5}{6}\right)^{50}$
	b		$H_0: p = \frac{1}{6}$ allow 0.167 or 0.17 $H_1: p < \frac{1}{6}$ $p = P(\text{dice shows a 2 (on any throw)})$ or $p = \text{proportion of 2s thrown}$	B1 B1 [2]	One error, eg undefined p or 2-tail or H_1 : $p > \frac{1}{6}$: B1B0 Two errors, eg $H_1: p > \frac{1}{6}$ and undefined p : B0B0 Cannot be scored in part (c)
	c		$B(50, \frac{1}{6})$ Attempt $P(X \leq a)$ for $1 \leq a \leq 20$ $P(X \leq 4) = 0.0643$ $P(X \leq 3) = 0.0238$ Rejection region is ≤ 3 twos or < 4 twos Alternative method ft if 12(b) $H_1: p > \frac{1}{6}$ $B(50, \frac{1}{6})$ Attempt $P(X \geq a)$ for $1 \leq a \leq 20$ $P(X \geq 14) = 0.0307$ $P(X \geq 13) = 0.0627$ Rejection region is ≥ 14 twos or > 13 twos	B1 M1 A1 A1 Alternative e method B1 M1 A1 A1 [4]	stated or implied or $P(X < 5) = 0.0643$ or $P(X < 4) = 0.0238$ Must see both probabilities correct oe. Allow $X \leq 3$ or $X < 4$ Alternative method ft if 12(b) $H_1: p > \frac{1}{6}$ stated or implied Must see both probabilities correct oe. Allow $X \geq 14$ or $X > 13$
			Total	7	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
2	a	Allow 2 sf throughout $H_0: p = 0.25$ where $p = P(\text{a packet contains gift})$ $H_1: p < 0.25$ $B(20, 0.25)$ & $X = 1$ $P(X \leq 1) = 0.0243$ comp 0.025 Reject H_0 Sufficient evidence that proportion containing gift is less than 0.25	B1(AO1.1) B1(AO2.5) M1(AO3.3) A1(AO3.4) A1(AO1.1) M1(AO1.1) A1(AO2.2b) [7]	or $p =$ proportion of packets containing gift One error, eg undefined p B1B0 soi Condone $P(X = 1)$ but not $P(X = 1) = 0.0243$ = 0.0211 or other incorrect dep 0.0243 and 0.025 Allow eg "H_0 is incorrect" Dep 0.0243 or $P(X \leq 1)$ stated or 0.0211 Can be implied by correct conclusion as for A1 below In context, not definite, eg not "Proportion is less" Examiner's Comments There were some very good responses to this question. However, it was clear that the candidates fell into two classes - those that had been trained to answer hypothesis test questions and those that had not been so well trained. A common error was to omit to define p in the hypotheses. Another, more serious, error was to find $P(X = 1)$ rather than $P(X \leq 1)$. A third common error was to give a definite conclusion such as "The proportion of packets containing a gift is less than 25%", rather than a correct, more limited, statement such as "There is significant evidence to suggest that the proportion of packets containing a gift is less than 25%".

Mark Scheme

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	b		EITHER whether a packet contains a free gift is not independent of whether other nearby packets contain the free gift OR eg The probability that a packet contains a gift is not the same for each packet or The proportion of packets with gifts in each box is not constant OR Free gifts not distributed randomly	B1(AO3.5b) [1]	Allow The probability of packet containing a gift is not independent Explanation, in context of why either the independence condition or the constant probability condition is not met. NOT The number of gifts in each box is not constant
			Total	8	

Mark Scheme

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3	a		e.g. The probability of side effects is the same for each patient (or is constant) e.g. The patients (or cases) are independent oe e.g. Patients form random sample of the population	B1(AO1.2) [1]	oe, in context Allow any indication of context, eg "patient" or "side effects" NOT Prob of patients getting side effects is indep or random. Ignore all else	<u>Examiner's Comments</u> Many candidates attempted to give an assumption in context referring to, for example, other conditions that might cause side effects, or patients being in the same hospital, or there being only two outcomes: side effects or no side effects. These generally did not address the main point. Many others addressed the main point, but did so just by quoting by rote from the text book, without reference to context. The mark was gained by a statement equivalent to one of the following. "Whether a patient experiences side effects is independent of whether any other patient does so." or "The probability that a patient experiences side effects is the same for each patient." or "Patients form a random sample from the population of people with the disease." Many gave roughly the following statement: "The probability that a patient experiences side effects is independent of other patients." This is not quite correct since it is the events that need to be independent, not the probabilities. Some candidates referred to the probability of a patient getting the disease, rather than experiencing side effects. Some candidates used the words constant, random and independent but not in the correct way. Some candidates gave assumptions that were possibly relevant to the experimental

Mark Scheme

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					design, but not relevant to the binomial distribution in particular, for example: We need to assume that the patients are telling the truth, and we need to assume that it is the medication that is causing the side effects. It is worth noting that two of the conditions for a binomial distribution (Repeated trials and only two possible outcomes to each trial) are not "assumptions" that need to be made in a particular context. They are conditions that are built into the context.	
	b		$H_0: p = 0.097$ where p is the proportion of patients experiencing side-effects (within a year) or 9.7% of patients experience side-effects $H_1: p > 0.097$ or $> 9.7\%$ exp side-effects $B(450, 0.097)$ & $X = 50$ (allow 51) $P(X \geq 51) = 1 - 0.862 = 0.138$ (3 sf) Compare 0.1 Insufficient evidence to reject H_0 Allow "Not reject H_0"	B1(AO1.1) B1(AO2.5) M1(AO3.3) A1(AO3.4) A1f(AO1.1) M1(AO1.1)	Allow "possibility" or, "probability" Not p = percentage having disease One error, eg undefined p, $p = 9.7\%$ B1B0 stated or implied eg by 0.138 or 0.107 or 0.108 or 0.862 or 0.893 or 0.0308 or 0.0366 NB 0.138 seen (or 0.107 or 0.108 or 0.862 or 0.893) implies M1 even if part of incorrect statement, eg $P(X \leq 51) = 0.107$ or $P(X \geq 51) = 0.138$ cao BC Dep 0.138 or 0.107 or 0.108 only Must see this statement oe Dep 0.138 or 0.107 or 0.108 or $P(X \geq 51)$ or 50)	If 2-tail test: $H_0: p = 0.097$ (defined p) B1 $H_1: p \neq 0.097$ B0 M1 A1 Comp 0.05 A1 No more marks Condone "Reject H_1" Might be implied by conclusion NB possible

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			Allow Accept H_0 No evidence (at 10% level) that proportion experiencing side-effects in one year under new treatment is greater than under standard treatment	A1f(AO2.2b) [7]	stated Any equivalent statement, in context, not definite, eg allow "likelihood", "percentage" Ignore all else ft only their $P(X \geq 51)$ and NB possible opposite conclusion on ft 0.108 seen: max B1B1M1A0A1M1A 1 Bin(450, 0.1) used: max B1B1M0A0A0M1A 1	opposite conclusion on ft NOT eg "change" NOT "proportion experiencing side effects is not significantly greater" Context can be implied by, eg, "patients", "proportion"
					Examiner's Comments Many candidates had clearly been well prepared for hypothesis test questions. However, a few candidates appeared to have no familiarity at all with this topic. Many candidates lost a mark because they correctly stated the hypotheses in terms of p, but did not define p. A few gave incorrect forms of the hypotheses such as $H_0: P(X = 0.975)$ and $P(H_0 = 0.975)$. A few used 0.1 in their hypotheses instead of 0.097. Most candidates used the correct binomial distribution and some stated that they were finding $P(X \geq 51)$. However, the calculations that many carried out were incorrect. Many actually calculated $1 - P(X \leq 51)$ or $P(X \leq 51)$ or $P(X \leq 50)$ or $P(X = 51)$ or $P(X = 50)$. Most candidates gave a conclusion referring to H_0 and a conclusion in context, and many of these were correct. Some	

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					candidates gave their conclusion as "There is evidence to accept H_0" or "There is evidence to reject H_1" rather than "There is insufficient evidence to reject H_0". These forms of the conclusion are not quite correct. Some gave the opposite to the correct conclusion. For example: "$0.138 > 0.1$ so reject H_0". A few compared their calculated probability with 0.097 instead of 0.1. A few gave their contextual conclusion in a definite form, for example "The proportion experiencing side effects under the new treatment is not greater than under the standard treatment." These lost the final mark.  AFL A common issue in hypothesis tests is candidates not securing the final marks for their conclusion, which need to be non-assertive and in context. Spec reference 2.05a gives examples of how the conclusion should be worded.
			Total	8	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
4		a	$P(X=3) = {}^5C_3 \times \frac{100}{300} \times \frac{99}{299} \times \frac{98}{298} \times \frac{200}{297} \times \frac{199}{296}$ $= 0.164318883 = 0.164 \text{ (3 sf)}$	M1(AO 1.1a) A1(AO 1.1) [2]	or equiv methods	
		b	P(disc is black) changes each trial (because no replacement) oe But change in prob is small oe Hence bin gives approx, but not exact, probs oe	E1(AO 2.4) E1(AO 2.4) E1(AO 3.5b) [3]		
		c	$P(X=3 \text{ using bin}) = {}^5C_3 \times \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^3$ $ (= 0.164609053)$ $\frac{'0.164609053' - '0.164318883'}{'0.164318883'} \times 100$ $= 0.177\%$	M1(AO 3.4) A1(AO 1.1) [2]	ft their values for M1 cao	
		d	$\mu = 1000 \times 0.164609053 (= 164.609053)$ $\sigma^2 = 1000 \times 0.164609053 \times (1 - 0.164609053)$ $ (= 137.5129127)$ $X \sim \text{Normal}$ $164.609053 \pm \sqrt{137.5129127}$ $= 152.88 \text{ to } 176.34$ Estimated limits are 153 to 176	M1(AO 3.3) M1(AO 1.1a) A1(AO 1.1) [3]	both np and npq correct method ft their μ and $\sqrt{\sigma^2}$. Allow rounding to 3 sf Allow (150 – 155) to (174 – 180)	ft their 0.164.6..... Allow use of 0.1643.... $165 \pm \sqrt{138}$ or better
			Total	10		

Mark Scheme

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5		a	i) $B(30, \frac{1}{6})$	B1(AO3.3) [1]	BC	
		a	ii) 0.0309	B1(AO1.1) [1]		
		a	iii) $1 - P(X \leq 8)$ = 0.0506	M1(AO3.4) A1(AO1.1) [2]		
		b	$H_0: p = \frac{1}{6}$ where $p = P(\text{score is 6})$ $H_1: p > \frac{1}{6}$ $P(X \geq 9) = \text{'0.0506'}$ or their (a)(ii) comp 0.05 Not reject H_0 No evidence that dice biased towards 6	B1(AO1.1) B1(AO2.5) B1(AO3.4) A1(AO1.1) M1(AO2.2b) A1f(AO3.5a) [6]	Undefined p : B0B1 BC ft their (a)(ii) ft their (a)(ii) In context, not definite	dice is unbiased B0 dice is biased towards 6 B1
			Total	10		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
6		a	$y^7 + 7xy^6 + 21x^2y^5 + 35x^3y^4$	B2 (AO1.1 1.1) [2]	B1 for three terms correct
		b	$21x^2y^5 = 35x^3y^4$ $\frac{x}{y} = \frac{3}{5}$ or 0.6	M1 (AO3.1a) A1 (AO1.1) [2]	Equate their terms in x^2y^5 and x^3y^4
		c	$P(L=2) = {}_7C_2 \left(\frac{3}{8}\right)^2 \left(\frac{5}{8}\right)^5$ and $P(L=3) = {}_7C_3 \left(\frac{3}{8}\right)^3 \left(\frac{5}{8}\right)^4$ Let $p = \frac{3}{8}$ and $1-p = \frac{5}{8}$, then $\frac{p}{1-p} = \frac{\frac{3}{8}}{\frac{5}{8}} = \frac{3}{5}$ so from part (b) $21p^2(1-p)^5 = 35p^3(1-p)^4$ and $P(L=2) = P(L=3)$	B1 (AO3.3) M1 (AO3.4) E1 (AO2.1) [3]	Connect to part (b) AG
			Total	7	